Sentiment Analysis of Dialogues: A Performance Comparison of ELECTRA, RoBERTa, DistilBERT, DeBERTa, BART, and SVM

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Abstract

Sentiment Analysis on Titanic Movie Dialogues This project works with the challenge of performing sentiment analysis on conversational data extracted from the Titanic movie. Using a dataset based on character dialogue our motive is to classify the emotional distribution of these conversations into positive, negative and neutral. Using pre-trained Hugging Face transformers pipeline we labelled the sentiment first then decided to normalize and form them as numerical labels. The dataset was then pre-processed for machine learning such as train-test split. The extensive scoring over a broad range of machine learning and deep learning techniques. We started with more traditional machine learning methods Support Vector Machines (SVM) with different kernels which gave accuracies of 64.42% (Linear), 59.23%(RBF), and 53.85% (polynomial), and 65 % in Voting Classifier . By doing on this base Our project can detect state-of-the-art transformer-based models for their own superior contextual understanding. In or project the DistilBERT gets the highest performance it has achieved an accuracy of (0.84) with 3 epochs, and then our second best model is RoBerta with the accuracy of (0.825) with the 5 epochs , and ELECTRA achieved 0.821 with 7 epochs . on the other hand BART models peaked at 0.798 accuracy with (5 epochs) , DeBERTa achieved its best at 0.810 (5epochs). we can clearly see the results highlight the advantage of transformer architectures in sentiment within the informa and varied language of movies dialogues, easily outperforming traditional SVM methods, DistillBERT come up with the most effective model for the task, and gives the most accurate solution for the sentiment classification on data(Titanic movie).The dataset presents challenges such as informal language,

varying sentence lengths, and context-dependent expressions, which make sentiment classification more complex.In this context, pre-processing modules like tokenization, padding, and attention masking proved helpful. Hyperparameter tuning was also done to maximize performance across architectures. To complement accuracy, more comprehensive evaluation of each architecture was done using precision, recall, F1-score, and accuracy. As seen from the results, the universal transformer-based models have relative advantages over models rooted in traditional machine learning in the context of handling contextual dependencies. The current work also demonstrates the value of using pre-trained embeddings as this reduced overall training time and increased pre-trained models' efficiency. The study also tends to show that transformer-based models exhibit advantages over traditional machine learning architectures in conversational contextual dependency tasks. The work sheds light on contextual understanding in conversational sentiment analysis tasks. There are a number of possible avenues to add value to this work in the future. One important area would be a more clear use of larger and more varied dialogue datasets to model to the point of having a truly generalizable model. Another area would be using different types of data like audio and visual cues in a multimodal framework to improve the overall accuracy of the use case.

Keywords: Sentiment analysis,Affective computing,Emotinal intelligence,Social networks,conversational data

1 Introduction

Sentiment Analysis belongs to Natural language processing (NLP) by using this process we can easily identify emotions and we can easily extract emotions from the given text. Put simply, it basically tries to identify whether a text is positive or negative or neutral. In this growing world where most of our communication is doing online(for example- texting,social media) a large amount of text data are created every second and automatic sentiment detection can help to understand public opinion, user behavior, and how we communicate.In our project, sentiment analysis is applied to a specific and interesting type of data: movie dialogues from Titanic. Unlike structured or formal writing, movie dialogues are conversational and deep. we have focus on these dialogues, our goal is not just to label them as positive or negative, but to better understand how emotions evolve throughout the storyline and how characters express them during different moments of the film.Sentiment analysis of conversations adds increased dimension compared to typical text analysis. Because conversations are more emotional and encode dimensions like sarcasm, and emotional dissonance and tension, they are more challenging to rationalize. In the context of the Titanic, studying sentimental irregularities and patterns convey how the audience's perception of each character evolves with the development of their emotional relationships and the character-building process of each scene. Other than film analysis, this technique can also help to develop more humanlike emotional response systems for conversational AIs.The analysis has been done according to this plan. The data setup was done with

the docker containers and the Titanic movie data set, which also has lines of character dialogues.Cleaned and organised data to fill in missing values: before applying any type of models, we had first to clean the data for it to be consistent. A high level sentiment classifier was run using a transformer pretrained via Hugging Face. These models, which are trained on very large datasets, understand context and linguistic features surprisingly well, and therefore represent a great ground starting point for topic and sentiment label allocation to every dialogue.The only drawback is that those models outputs are not always in the same format or prediction labels To mitigate this, the outputs are rounded up to 3 categories: POSITIVE,NEGATIVE and NEUTRAL This acts as a step of normalization to make sure all data points follow the same labeling scheme, which eases further analysis.After that all the categorical labels was converted by us manually into numbers using LabelEncoder from sklearn. This step is important to training since machine learning models work on numbers and not text labels. Here, we used the dialogues as the input features(X), and encoded sentiment labels as target variable(y). Affective computing, This branch of computing was pioneered by Rosalind Picard and concerns the building of computer systems that understand human emotions and can react to them. Compared to traditional computing, which uses logic and cycles through structured inputs and outputs, this branch uses the the most basic, basic, less structured shape of reality, and that is how people feel. It integrates into the approaches of computer science, psychology, cognitive science, and linguistics to reduce the gap between how machines think and how people feel. In many tier if we think on basic tier also (ex-sentiment analysis) the systems, or simply software, classify texts by degrees, or levels, of positive/negative/neutral. It is relatively useful, but at the same time, limited. Humans are far from simple. Our vision of affective computing is broader, aiming to not only classify and discern the feelings of people and which emotions dominate (i.e., is it fear/anger/joy/sadness, and of what tier tier)and is it sadness or tier tier shad. Sadness and despair, though minor, they are not at the same level. This is affective computing, the science of computing that perceives these nuances. This is even more applicable to dialogue systems and conversational data, not excluding the quarreling, the back-and-forth naturally occurring in human discourse, or even the largely prescriptive dialogue present in movie scripts, like the scripts of the movie Titanic. Scripts of movies are much more than sequences of characters—they are manifestations of a particular emotion in a temporal and dynamic shape. They also play an integral and foundational role in telos and character motivation and development. A basic sentiment analysis might suggest lines are "negative," but that doesn't tell the analyst if the character is, say, afraid, or if they are grieving or seething with anger. A superior and affect-aware computing system might notice these nuances, and thereby afford us an even more sophisticated understanding of character motivation.Affective computing encompasses more than just the analysis of text. Things like tone of voice, facial expressions, and body language also help achieve text analysis. We have a basic understanding of what emotions are in this project. Affective computing's main aim is to create systems that [insample text] emotional *intelligence*. As you all say, the systems do not *feel* emotions. The *recognition* and *response* of things that concern the emotions are what matter. From the practical point of view of this development, the result will be more *natural*

and *realistic emotional* than interactions in chat and voice systems. When working with a current project, more tool-sentiments will lose the surface of the emotional diagnostics that are meant to progress the systems from the guessing verbal content and its emotional significance.

Social networks have fundamentally changed the way humans interact, to the point where they are a ubiquitous part of everyone's daily lives-the place where we share opinions, converse, and express feelings. Hundreds of millions of users express themselves on social networks every second of the day; and this creates a huge amount of data that is intrinsically rich in content and emotional state. This data stream, known as social network data or social media data, is a dynamic, instantaneous, and relatively unfiltered stream of human communication. From the perspective of both sentiment analysis and affective computing, conversational data found on these platforms (and, by extension, structured narrative conversation like movie dialogue) offers a window into collective, as well as individual, affective state.Social network conversations come in many forms including posts, comments, replies, direct messages, and group chats; and when gathered, these forms create vast streams of data that we can examine for public opinion on every subject, from popular brands, to political figures, social issues, and global affairs. The real-time nature of social media allows for instant capture of public feeling, and thus, provides insight into everything from crisis management to trend prediction and marketing research. Yet the richness of social network data also comes with considerable analytical complexity; due to the relatively unstructured nature of conversation on these networks, these conversations are full of slang, acronyms, emoticons, memes, and colloquial speech. This complex mix severely complicates machines' ability to identify emotions. Such concepts as sarcasm and irony in conversations make these tasks additionally difficult as they require a user to interpret language based on the contrary of what they actually read.Dialogues extracted from the Titanic movie could be considered a subcategory of conversational data similar to that found on social networks: conversations, though curated, structured, and designed to tell a story, still represent human discourse that is meant to express emotion and carry meaning and intent. The study of emotions expressed and identified in movie dialogues can inform models for the understanding and recognition of emotions in other, less structured data streams. This project focuses on dialogues of the Titanic movie as a particular form of this conversational data and examine emotions conveyed throughout the film.The large size and high velocity of social network data streams, as well as their inherent linguistic complexity, necessitates computationally powerful methods. Traditional text analysis cannot easily cope with the scale and variability of data of this type, thus explaining why a transformer-based deep learning model (e.g., BART, DeBERTa, DistilBERT, ELECTRA, RoBERTa) would perform well on this data: their inherent structure and understanding of semantic contexts within a piece of text allows for a greater level of accuracy in parsing and processing informal text than traditional models. By studying the transmission of emotions from one character to another in a controlled environment such as a movie script, a more general understanding of emotional interaction can be formulated and expanded.Combined with affective computing methods, analysis of these data streams

from sources like the Titanic movie offers more than simple sentiment classification-instead of detecting negative or positive sentiment associated with dialogue regarding the film's demise, an affective computing model would be able to categorize emotions into much finer states such as sadness, fear, anger, and despair. Such detailed classifications not only allow for deeper understanding of characters, but also of human expression of emotion. Research into the way characters interact affectively throughout this movie gives insight into both film studies and NLP as well as advanced AI methods of modeling human emotion. To go beyond basic categorizations of positive, negative or neutral sentiment, and truly explore the richer and more nuanced world of individual emotions, a strong theoretical underpinning is necessary. One of the most well-known and useful psychoevolutionary models of the complex array of human emotions is Robert Plutchik's Wheel of Emotions. Published in 1980, Plutchik theory categorizes eight primary or basic emotions; Joy, Sadness, Trust, Disgust, Fear, Anger, Surprise and Anticipation. Plutchik posited these primary emotions as evolutionarily primitive and biologically innately programmed states that help organisms survive by prompting adaptive behaviors in response to specific environmental stimuli.One of the most powerful aspects of Plutchik's model is the way it is structured hierarchically and dimensionally, usually presented visually as a colorful wheel or as a three dimensional cone. Through these ingenious structures, several characteristics and relationships between emotions are represented. Intensity: Emotions are not of fixed value but exist on a scale of intensity. At the core of the wheel are the most intensely felt emotions, with fainter states at the outer edges of the wheel. Annoyance is a low-intensity form of Anger, just as ecstasy is a high-intensity form of Joy. The intensity of the expressed emotion is crucial information in affective computing since knowing a speaker or character's emotional state at a higher fidelity of specificity will lead to more accurate and insightful interpretations. Understanding that a character in Titanic has a state of "Apprehension" as opposed to "Terror", for example, can develop one's knowledge of the character, and the level of tension and understanding of the character. Polarity: Plutchik's wheel places opposite emotions on opposite sides of the wheel. This includes sadness vs. joy and fear vs. anger. This can be an efficient way to chart contrasting emotional states. Mixtures (Dyads): just how primary colors can be combined to create different colors, primary positions can be synthesized to create secondary positions that can be called Dyads.Joy can be blended with trust to create Love, while surprise blended with sadness creates disappointment. Identification and analysis of these complex blended emotions is essential for interpreting human speech, as language is rarely mono-semic but always a complex blend of individual emotion states. Understanding that Titanic dialogue is, say, "anxiety" (fear + anticipation) or "remorse" (sadness + disgust) rather than "negative," brings much greater clarity to the expression. When using Plutchik's model to analyze conversations, particularly those from the Titanic movie, the ability to transcend generalized sentiment becomes incredibly valuable for a much deeper and more precise understanding of emotional state. Instead of simply understanding a character's dialogue to be "negative", an affective computing system, working with Plutchik's framework, could perhaps even identify it as expressing "anger," "fear," "sadness," or "disgust"

and potentially "contempt" (a mixture of Anger and Disgust) or "submission" (a mixture of fear and trust). in narrative settings such as film.In analyzing the dramatic discourse that takes place on the Titanic, understanding that a character expresses "grief" (an intense form of sadness) or "terror" (an intense form of fear) for the imminent disaster on the ship brings much more meaning to the emotional journey than a simple "negative" label could ever provide. Although our initial classification outputs for Titanic dialogues simply label expressed sentences as POSITIVE, NEGATIVE, or NEUTRAL, this project represents an attempt to get to the deeper and richer end of affect and understand emotion through the more comprehensive model Plutchik offers, which can form a basis for even richer future systems and labels than that presented.

This study focuses on the sentiment analysis of movie texts through the lens of both traditional machine learning and transformer-based models. Models such as BART, RoBERTa, DeBERTa, DistilBERT, and ELECTRA are used to assess context and semantics of movie text. Their performances are evaluated over many training epochs and are contrastively evaluated against Support Vector Machine (SVM) classifiers of various kernels. The purpose of the study is to assess and compare the effectiveness of models using commonly accepted sentiment analysis metrics such as accuracy, precision, recall, F1 score, and Matthews Correlation Coefficient (MCC). The effectiveness of the various approaches to sentiment analysis are discussed through the results.

2 Literature review

Kim (2015) stated that Convolutional Neural Networks (CNNs) can be effectively applied to sentence classification tasks, including sentiment analysis. The study demonstrated that CNN models, when combined with pre-trained word embeddings, are capable of automatically extracting meaningful textual features without the need for manual feature engineering.This method had a significant representation of cross-section across various benchmarks, proving the advantages of DL methods over ML approaches. Nevertheless, the study didn't cover the challenges of CNNs relying on localized components and on the inability to capture text's extensive range components. Moreover, the model fails to achieve deep contextual comprehension and bidirectional processing, which are vital to comprehend the intricate, complicated, and delicate structures of various sentences.

Tang et al. (2015)showed that traditional sentiment analysis approaches struggle to recognize sentiment in context, particularly in short, succinct formats, such as tweets. To fill the gap, the authors proposed a sentiment-sensitive word embedding approach channeling word representations through sentiment in the embedding space. This way, the proposed method optimally performs sentiment classification tasks by producing embeddings that are sensitive to polarity of sentiment in comparison to standard embeddings. Nonetheless, the authors could not deal with the complexity of the context of longer sentences. On the downside, the method relies on sentiment indicators that are structured in a training phase, which would hinder the method's applicability in a multi-domain approach. The proposed method also lacks depth of understanding in the context compared to the transformer based models that came later.

Zhang et al. (2015)reported that Convolutional Neural Networks (CNN‘s) that focus at a character-level rather than a word-level have advantages when applied to the task of text classification. The study showed Character-based models as a technique that has the ability to scale to large-structured data sets and learn features about the text through the use of the deep learning technique.However, the model requires a large amount of training data and is computationally expensive, making it less efficient for smaller datasets.

dos Santos and Gatti (2015) stated that deep Convolutional Neural Networks can be applied for sentiment analysis at both sentence and document levels. Their approach incorporated word-level embeddings and achieved improved performance over traditional machine learning methods. However, the model still struggles with capturing long-range dependencies and lacks deeper contextual understanding of complex sentences.

Tai et al. (2015) stated that Tree-structured Long Short-Term Memory (Tree-LSTM) networks can improve sentiment analysis by modeling hierarchical sentence structures. The study showed that incorporating syntactic structures enhances the model's ability to understand sentence-level sentiment. However, the approach depends on accurate parsing and increases computational complexity, making it less scalable for large datasets.

Joulin et al. (2016) stated that efficient text classification can be achieved using a simple and scalable approach called FastText. The study presented the idea of text represented by bag of words or n-grams and then classified with a linear classifier. This approach was found to reduce training time and shows comparable accuracy. The model performed well on large datasets and was also efficient in terms of computation unlike the deep learning tools. However, the deep learning tools model a language more deeply compared to this approach. This also allows the tools to have a better grasp of the deeper semantics of language. This is crucial in the field of sentiment analysis to understand the subtle nuances of text. The approach outlined in the study is therefore not ideal for sentiment analysis tasks.

Zhang et al. (2016) Deep learning models like Convolutional Neural Networks can be enhanced by modifying architecture depth and feature extraction dynamics. They posited that deeper CNN models can move toward performing better on text classification due to their ability to capture more intricate data patterns. Although deeper models can classify complex data more effectively, the model still does not perform well on extracting long-range patterns or contextual data. In addition, larger models require more computational resources and result in longer training times.

Yang et al. (2016)Hierarchical Attention Networks (HAN) can boost document classification by evaluating the importance of words and sentences. Deciding what part of the text to apply attention to improves the model's performance and explainability. Although the model has enhanced the interpretability of text document classification, it is computationally expensive and fragile to train. Additionally, the model's dependence to a sequential process limits the model's capability to train in parallel.

Johnson and Zhang (2016) said that CNN models that use region embedding can enhance text classification by learning textual representations of regions rather than words. In this case, the approach showed better accuracy than standard CNN models.

Nevertheless, the approximate method still provides limited contextual understanding, focusing on region-related representation while neglecting global dependencies in the text.
Ribeiro et al. (2016)said model interpretability is an essential part of any machine learning system. To this end, LIME, (Local Interpretable Model-Agnostic Explanations) is one such model aimed at explaining the predictions of black-box models. LIME is one such method that improves interpretability by providing insight into the significance of a feature for a given instance prediction. Despite being a method that explains black-box models, LIME may create representations that approximate local model behavior and can create representation explainability's uncertainty.
Vaswani et al. (2017) said traditional sequence models like RNNs and CNNs are unable to capture long-range interactions, and are unable to do computations in parallel. In their work, a new type of model, the Transformer architecture, primarily one that is based on attention, with no recurrence or convolution, is proposed. That model showed state of the art results on machine translation and also allowed parallelization. Unlike the aforementioned models, the transformer model is resource and data scarce.
Peters et al. (2017) said the use of contextualized word representations for improving the results of many NLP tasks is significant. Through the use of ELMo, one of a deep contextualized word representations model, the impact of the entire context (i.e. the sentence) is acknowledged. Consequently, the performance of various models is improved on a range of tasks. Unlike the work of previous models, ELMo is also resource scarce.
Lin et al. (2017)stated that implement self-attention mechanisms, and utilize focus in several regions in a sentence. Besides improving sentence representation, the model improves interpretability and performance on classification tasks. Yet, it adds complexity and requires delicacy and precision when setting attention levels.
Ribeiro et al. (2017)stated interpretability continues to be one of the biggest hurdles in the realm of machine learning. They developed techniques focused on explaining the predictions in the realm of complex classification. This study improves transparency and builds trust. Nonetheless, these explanations continue to be local, and fail to show the behavior of the entire model.
McAuley et al. (2017)stated neural models are applicable to sentiment and recommendation tasks by learning both textual data and user-item relations. The method improves both personalization and prediction, but lacks efficiency on fully utilized graphs and requires high levels of computation.
Devlin et al. (2018)stated that to a significant extent, understanding language requires context to manifest in both directions. They introduced BERT (Bidirectional Encoder Representations from Transformers) and explained that the architecture of the transformer makes it possible to learn deep representations that are both bidirectional. The model delivered top-of-the-line performance in many language tasks. But, the model itself is costly, and requires a considerable amount of data in the form of pre-training.
Radford et al. (2018)stated that generative pre-trained transformer (GPT) is capable of achieving good performance in many language tasks by performing a combination of unsupervised pre-training and then fine-tuning. Bessifier and Lesgardieux explain

that the model faces a significant challenge in that it is unidirectional, and lacks understanding of full bidirectional context.
Howard and Ruder (2018)stated that transfer learning mechanisms when applied in the context of NLP positively impact performance. They presented ULMFiT as a method of fine-tuning on the pre-trained model of language for specific custom tasks. The method is both efficient and applicable for smaller datasets. However, it still depends applicably on sequential processing, and as a result, fails to capture full contextual relations like a transformer.
Ribeiro et al. (2018) emphasizes the importance of reliability in interpretability techniques. They propose improvements in explanation methods focused on trust and usability. Despite progress, explanations are still approximate and may fluctuate with different inputs.
Cer et al. (2018) argues that all-encompassing, universal sentence embeddings can enhance transfer learning for all NLP (Natural Language Processing) tasks. They developed the Universal Sentence Encoder, which creates fixed-length sentence embeddings. Still, the model may struggle to capture the gradual and intricate contextual details of deep sentences.
Liu et al. (2019) claims the precursor for pre-trained models of BERT has great potential for significant improvements. They developed RoBERTa, which removes next sentence prediction and only trains on more significant samples (i.e. the more lengthy sequences). The model achieved top results on many of the evaluations. Unfortunately, balancing minimal resources (computing) and optimal performance is challenging.
Yang et al. (2019) predicts that trade-offs from permutation-based language modeling can provide (short-term) contextual advantages. They developed XLNet, which samples bidirectional context (eliminates the need to mask). XLNet improved performance of BERT in several tasks of NLP. Still, XLNet brings (short-term) increase in computation and (short-term) increase in training time.
Sanh et al. (2019) claims the performance of large transformer models can be increased with significant compression. They developed DistilBERT which is a minimal (smaller) version of BERT. Throughout the performance optimization process of spanning BERT, DistilBERT advocates for the optimization process to be as (short-term) "minimal" as possible. DistilBERT is able to (short-term) be faster and (short-term) smaller as BERT.
Sun et al. (2019) claims that for sentiment analysis, the (long-term) outcomes can be improved with the fine-tuning from the pre-trained models (Transformer models). The results is evidence that (sentiment) fine-tuning from frameworks (long-term) can be effective. Unfortunately, balancing optimal performance and minimal resources (computing) is challenging.
Lundberg and Lee (2019) function at a more advanced level, where they can utilize context, train them with fewer parameters, and require less compute. They built DeBERTa, utilizing disentangled attention and improving upon BERT and RoBERTa. DeBERTa provided more separation between positional and content information. Increased performance with more compute and more complexity was provided.

He et al. (2020) suggested a framework. He built DeBERTa, a more flexible mechanism at the level of the framework but still contextually aware, efficient, and still maintaining a fixed position and more providing content information. DeBERTa provided more separation between positional and content information. Increased performance with more compute and more complexity was provided.
Raffel et al. (2020) after they observed that the pre-training, finetuning, and inferencing of language models were highly costly, inefficient, and in turn, wasting a lot of compute and time. They built ELECTRA, a more computationally efficient model that presented similar performance with less pre-training and compute. Increased performance with more compute and more complexity was provided.
Clark et al. (2020) positive reception of T5, they concluded that the state of the art (SotA) in efficient NLP modeling was optimal with having a model that was only catered to the generation of text, thus both of training and fine-tuning have to be done to fit generations. Increased performance with more compute and more complexity was provided.
Brown et al. (2020)from the reduced fine-tuning ratio of models. They were able to rationalize the stacking of multiple autoregressive and highly complex models with the same frameworks, thus became their own frameworks that were able to be manipulated. Increased performance with more compute and more complexity was provided.simplified modeling with generation. Basic SotA performance on multiple generations. Increased performance with more compute and more complexity was provided.
Samek et al. (2020) said that to understand deep learning models, you have to have explainable AI. The study analyzed several XAI approaches, focusing on the value of explainability in decision-making systems. Still, only some of XAI methods are reliable, and they may not always generalize to other systems.
Fedus et al. (2021) said that using sparse activation to scale models supplement transformer models. For example, they have the Switch Transformer, which is a mixture of expert systems. The architecture scales models and token computation is minimized. They also maintain their efficiency done. It also has strong performance done, but at the cost of training instability; therefore, experts' balancing is needed.
Lewis et al. (2021) said that the fusion of retrieval and generative systems has a positive effect on factual and knowledge performance. Retrieval-Augmented Generation (RAG), is an example of using external knowledge to transformer systems. Knowledge-heavy performance then is done at the expense of system performance and reliance on documents.
Beltagy et al. (2021) said that the standard transforme rs cannot manage long documents. For instance, the normalization of long documents, Longformer, deals with sparse attention mechanisms and elongations. Resource consumption and careful architecture normalization are also done optimally.
Doshi-Velez and Kim (2021) said that the transparency of implementations of AI systems using deep learning is crucial, and for the study, the researchers focused on the absence and need for transparency of XAI. To a certain extent, the approaches focus on certain XAI methods.

Tamkin et al. (2021) say reinterpretability of deep learning systems is of great importance, and the answers show deep learning systems are no exception.
Chowdhery et al. (2022) said scaling language models makes performance on tasks better. A large language reasoning and understanding model based on transformer architecture called PaLM was developed. It excels in these areas. It requires high amounts of computing resources and also brings concern regarding how it's biased and ethical.
Ouyang et al. (2022) stated something about language models being more useful when aligned with humans. They provide an example of InstructGPT and argue that more user-aligned and quality-received models, are improved in their user services. There are concerns on the quality and the expensive nature of the human feedback process.
Hoffmann et al. (2022) said something about how performance is only guaranteed by the optimal scaling of the model and data. An example is Chinchilla, which creates balance between model size and the data, to achieve better efficiency. Of concern, is the balance between the resources and the performance.
Wei et al. (2022) stated that large language models that use the chain-of-thought prompting to help them reason, perform better even given the nature of the task. There are concerns regarding the increase in costs and the potential for reasoning inconsistencies.
Rudin (2022)said something about the nature of high-stakes problems and the use of black-box models. The study says that black-box models become inherently making post-hoc interpretations. The balance between performance and complexity of deep learning models becomes concerned.
OpenAI (2023) Stated that GPT-4 can process both text and images, and has been reported to be a useful addition to tools that conduct complex reasoning. But, for some organizations, the tools it creates are too resource-demanding, and for many users it can be too resource-demanding to use. More so, it has the potential for generating biased information and hallucinations.
Touvron et al. (2023) introduced the idea that spanning fewer processing resources can result in fewer resource-demanding tools. tools that span fewer resources are less computationally expensive and can reach the same performance benchmarks as tools that are more resource-demanding. However, it is still expensive and resource-demanding.
Dettmers et al. (2023) The idea is that spanning fewer resources can result in fewer resource-demanding tools. They offered a research tool that aims to reduce the processing resources and demands to span a resource-demanding tool. But, the tools it creates have the potential to span fewer resources.
Schick et al. (2024) stated that greater efficiency can be achieved through the different ways a tool can be created. For example, more can be achieved through combining multiple tools into a system to achieve a desired goal than through a series of tools. However, spanning more tools can increase the likelihood of multiple errors in the system.
Bai et al. (2025) They offered that multi-stage tools that span a greater processing system can be more efficient and reachable goals. The multiple stages can span greater processing demands. However, the tools are not the only demand. Placing tools at the

center of a system (multi-stage) to span greater processing demands. But, to achieve greater resource-demanding goals, more tools is not the only demand.

Table 1: Literature Review Summary

Label	Category	Contribution	Limitation
1	Deep Learning	CNN-based model for sentiment analysis using word embeddings and automatic feature extraction.	Captures only local features and lacks long-range contextual understanding.
2	Deep Learning	Sentiment-specific word embeddings that incorporate polarity information into word representations, improving sentiment classification performance.	Limited contextual understanding and depends on predefined sentiment signals; struggles with complex sentence structures.
3	Deep Learning	Character-level CNN for text classification without relying on word-level features.	Requires large datasets and high computational cost.
4	Deep Learning	Deep CNN for sentence and document-level sentiment analysis using word embeddings.	Struggles with long-range dependencies and lacks deep contextual understanding.
5	Deep Learning	Tree-structured LSTM model that captures hierarchical sentence structure for sentiment analysis.	Depends on syntactic parsing and increases computational complexity.
6	Machine Learning	FastText model for efficient text classification using n-grams and linear classifier.	Lacks deep contextual understanding and cannot capture complex semantic relationships.
7	Deep Learning	Improved deep CNN architecture for enhanced feature extraction in text classification.	High computational cost and inability to capture long-range dependencies effectively.
8	Deep Learning	Hierarchical Attention Network capturing word and sentence-level importance.	Computationally expensive and limited parallelization due to sequential structure.
9	Deep Learning	Region embedding-based CNN improving text representation and classification accuracy.	Limited global context understanding and dependency modeling.
10	XAI	LIME method for explaining predictions of black-box models.	Provides only local explanations and may lack global consistency.
11	Deep Learning	Transformer architecture using self-attention for sequence modeling.	Requires large datasets and high computational resources.
12	Deep Learning	Contextual word embeddings (ELMo) capturing sentence-level meaning.	Computationally expensive and limited parallelization.

Label	Category	Contribution	Limitation
13	Deep Learning	Structured self-attention for improved sentence representation.	Increased model complexity and sensitivity to parameter tuning.
14	XAI	Improved techniques for explaining predictions of black-box models.	Provides local explanations and lacks global interpretability.
15	Machine Learning	Neural approach for combining text and user-item interactions in prediction tasks.	Requires large-scale data and high computational cost.
16	Deep Learning	BERT model using bidirectional transformer for contextual language understanding.	Requires large computational resources and extensive pre-training data.
17	Deep Learning	GPT model using generative pre-training for NLP tasks.	Unidirectional context limits full understanding of text.
18	Deep Learning	ULMFiT model enabling transfer learning for NLP tasks.	Limited contextual understanding compared to transformer models.
19	XAI	Improved interpretability techniques for enhancing trust in ML models.	Explanations are approximate and may lack consistency.
20	Machine Learning	Universal Sentence Encoder for generating sentence-level embeddings.	Limited ability to capture fine-grained contextual nuances.
21	Deep Learning	RoBERTa improves BERT through optimized training and larger datasets.	Requires high computational resources and large-scale data.
22	Deep Learning	XLNet uses permutation-based learning for better context modeling.	High computational complexity and slower training.
23	Deep Learning	DistilBERT reduces model size and improves efficiency using knowledge distillation.	Slight reduction in accuracy compared to full BERT.
24	Deep Learning	Fine-tuning BERT for improved performance in sentiment analysis tasks.	Requires careful tuning and high computational cost.
25	XAI	SHAP method for consistent and interpretable feature importance explanations.	Computationally expensive for large models.
26	Deep Learning	DeBERTa improves transformers using disentangled attention mechanisms.	Increased model complexity and high computational cost.
27	Deep Learning	T5 model unifies NLP tasks into a text-to-text framework.	Requires large datasets and high computational resources.

Label	Category	Contribution	Limitation
28	Deep Learning	ELECTRA uses discriminative pre-training for efficient learning.	Sensitive to hyperparameters and training configuration.
29	Deep Learning	GPT-3 enables few-shot learning with large-scale language modeling.	Extremely high computational cost and potential bias issues.
30	XAI	Overview of explainable AI techniques for interpreting deep learning models.	Lack of consistency and limited generalization across models.
31	Deep Learning	Switch Transformer scales models efficiently using mixture-of-experts.	Training instability and complex expert balancing.
32	Deep Learning	RAG combines retrieval with generation for better knowledge-based predictions.	Depends on retrieval quality and increases system complexity.
33	Deep Learning	Longformer enables efficient processing of long documents using sparse attention.	Requires tuning and still computationally intensive.
34	XAI	Framework for evaluating interpretability in machine learning systems.	Lack of standardized evaluation metrics for explainability.
35	XAI	Analysis of internal representations in transformer models for interpretability.	Interpretability remains limited and model-dependent.
36	Deep Learning	PaLM scales transformer models for improved reasoning and language tasks.	Extremely high computational cost and ethical concerns.
37	Deep Learning	RLHF-based InstructGPT improves alignment with human preferences.	Depends on human feedback and increases training cost.
38	Deep Learning	Chinchilla optimizes scaling between model size and training data.	Requires careful tuning and substantial resources.
39	Deep Learning	Chain-of-thought prompting improves reasoning in language models.	Increased inference cost and possible inconsistency.
40	XAI	Advocates for inherently interpretable models over black-box approaches.	May reduce performance compared to complex models.
41	Deep Learning	GPT-4 enhances reasoning and supports multimodal capabilities.	High computational cost and issues like hallucination and bias.
42	Deep Learning	LLaMA provides efficient large language models with fewer parameters.	Requires fine-tuning and large datasets.

Label	Category	Contribution	Limitation
43	Deep Learning	QLoRA enables efficient fine-tuning of LLMs using quantization.	Slight reduction in model performance.
44	Deep Learning	Tool-augmented LLMs enhance reasoning and factual accuracy.	Increased complexity and reliance on external tools.
45	XAI	Alignment techniques improve safety and reliability of LLMs.	Complete alignment with human values remains unresolved.

3 Methodology

3.1 Dataset

The Titanic screenplay, listed on IMSDB, was utilized to generate a custom movie dialogue dataset for this study. Due to the unavailability of a properly structured dataset for this particular purpose, the dataset was formed manually through a methodical extraction and preprocessing pipeline. The unprocessed screenplay was gathered as raw text, subsequently undergoing various text parsing methods in Python. While constructing the dataset, the script was scrutinized for dialogue and the characters who spoke them. This meant identifying names, which are customarily written in capital letters, and identifying generic conventions in screenplays. Each dialogue line was then associated with its character. Using rule-based methods, non-dialogue elements were filtered, ensuring that aligned with the meaningful conversational text. The created dataset was structured manually in a tabulated form, consisting of dialogue and character pairs. Further, additional improvements were made to the dataset quality through preprocessing. These improvements were the elimination of duplicates and incomplete entries, as well as the filtering of non-dialogue text, normalization, and text consistency. To achieve a balance between diversity and processing efficiency, the dataset was pruned and normalized to around 2600 entries. For the purposes of annotation, sentiment labels were automatically created using a transformer-based natural language processing model from the Hugging Face Inference API. The model utilizes contextual semantics to guide the annotation process as opposed to simple keyword matching. The model assessed each dialogue's emotional tone, intention, and linguistic context and assigned them a sentiment classification of positive, negative, or neutral. This method guaranteed consistent and scalable labeling for the entire dataset. Unlike human annotations that can take time and involve a lot of subjectivity, a pre-trained transformer model offers a more consistent way of classifying sentiment. The quality of annotation, though, relies on how the underlying model has been trained and may not reliably identify subtle or context-relevant emotions. The limitations notwithstanding, the dataset is contextually rich and realistic representation of conversational dialogue, making it fit for the purposes of assessing both classical machine-learning models and state-of-the-art transformer models.

3.1.1 Data Preparation

The dataset preparation process includes multiple steps: data extraction, data cleaning, data structuring, and data annotation.
Data Extraction - The screenplay for the movie *Titanic* was accessed through the Internet Movie Script Database (IMSDB) and parsed using a custom Python script to extract portions of text and annotate them with corresponding characters.
Data Cleaning and Structuring - The extracted script contained narrative descriptions, scene directions, and inconsistent formatting that were removed using rule-based text processing techniques. Only relevant dialogue lines were retained and assigned to the corresponding characters. Noise such as scene headers (e.g., "INT.", "EXT.") and non-dialogue text was removed. The structured data was then organized as character–dialogue pairs.
Dataset Filtering - Data quality and consistency were improved by removing irrelevant and ambiguous entries. The dataset was refined to approximately 2,600 rows to ensure a balance between variety and computational efficiency.
Sentiment Annotation - Sentiment labels were assigned using a pre-trained transformer-based model via the Hugging Face inference pipeline. Each dialogue was classified as Positive, Negative, or Neutral.
The Final Dataset - The dataset used for training and evaluating various machine learning and transformer models consists of structured entries containing the character name, dialogue text, and corresponding sentiment label.

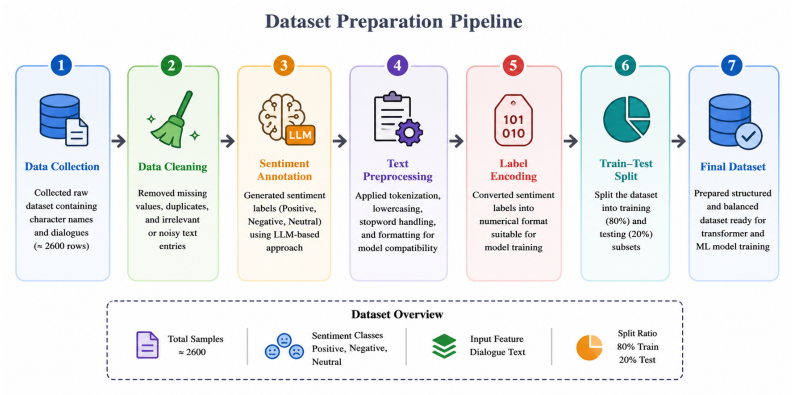

Fig. 1 Dataset Preparation Pipeline

3.2 Models

3.2.1 BART

BART (Bidirectional and Auto-Regressive Transformers) is a sequence-to-sequence transformer model that combines the strengths of BERT and GPT. It is particularly effective for text generation and understanding tasks due to its denoising autoencoder pre-training strategy.

3.2.2 DistilBERT

DistilBERT is a compressed version of BERT that retains most of its performance while being significantly lighter and faster. It is trained using knowledge distillation and is well-suited for efficient sentiment classification tasks.

3.2.3 DeBERTa

DeBERTa (Decoding-enhanced BERT with Disentangled Attention) improves upon BERT and RoBERTa by separating content and position information through disentangled attention mechanisms, resulting in better contextual understanding.

3.2.4 ELECTRA

ELECTRA is a transformer model that employs a replaced token detection approach as opposed to masked language modeling. This approach is both computationally efficient and effective for a variety of NLP tasks.

3.2.5 RoBERTa

RoBERTa is an optimized part of BERT trained with diff types of larger datasets.It also removes the next sentence prediction also and uses dynamic masking, improving overall performance.

4 Results

The results showed that the transformer works bettter than SVM techniques for labelling the sentiment analysis in dataset of titanic.Transformer models like Roberta distilBert Electra BARt can understand better because they look at the meaning of the sentences in the line and on the other hand SVm looks at words only thats why it performed better.so we have understood from this results that transformers models are better for doing sentiment analysis.

4.1 Performance of LLM

we have trained the BART model from the Large language models n we got a high accuracy of 0.798 at 5 epochs.It shows that the BART is good at understand n learning the context of the sentences for sentiment analysis(like positive, negative, neutral).

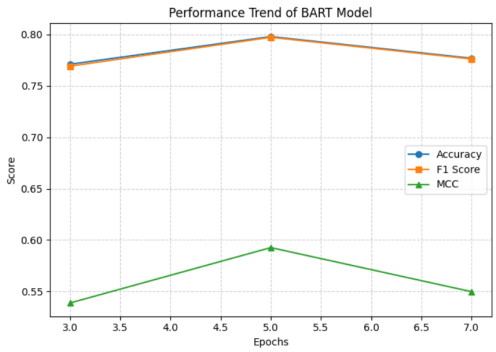

Fig. 2 Performance trend of BART model across different epochs

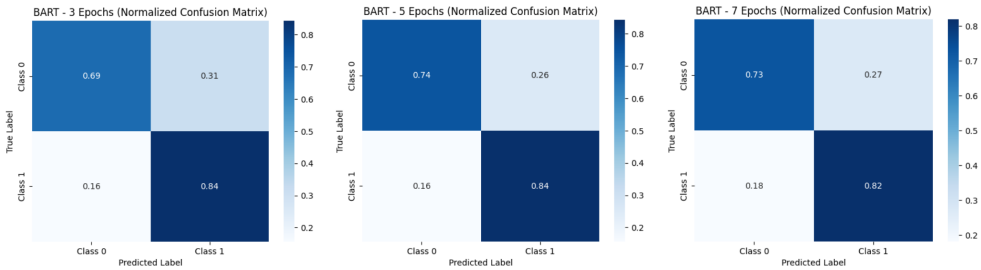

Fig. 3 Normalized confusion matrices for BART at 3, 5, and 7 epochs

4.2 Performance of Trannsformer

So from all the transformers models (like DistilBERt, ELECTRA, RoBERTa,DeBERTa) DistilBERT got the hiighest accuracy of 84.0% at just 3 epochs it shows good balance between both speed and performance.DistilBERT model is a good choice for sentiment analysis because it has good performance and the setup is also not hard.If we talk about RoBERTa it also has peformed very well, Its accuracy is 82.5% after 5 epochs, which shows that it has been trained effectively. ELECTRA also achieved similar results, with an accuracy of 82.1% after 7 epochs.

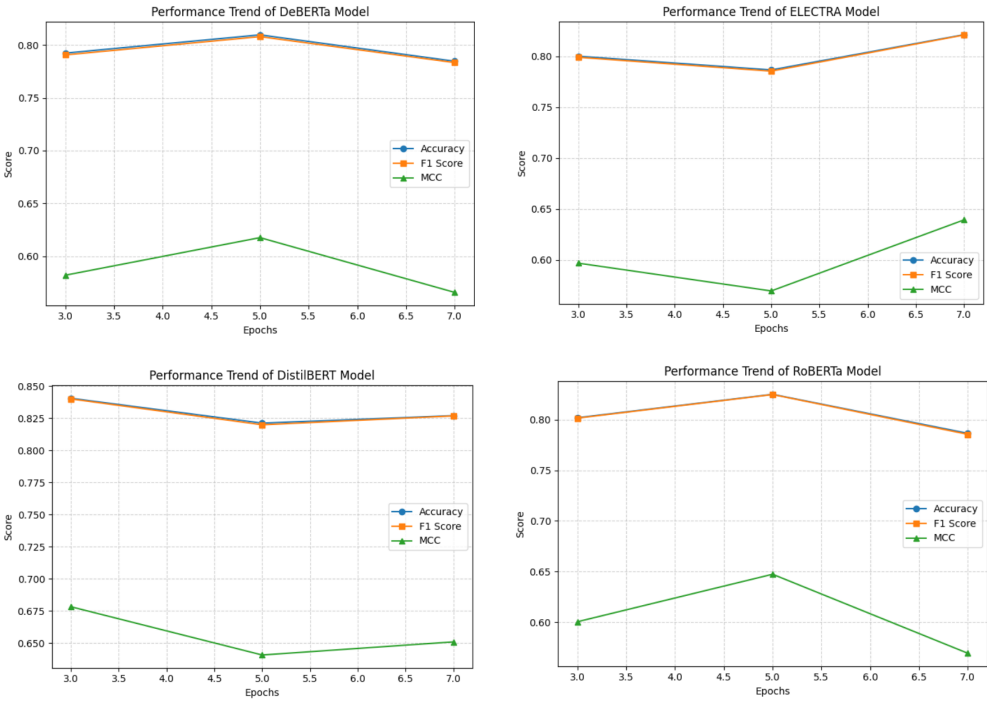

Fig. 4 Comparison of models: (a) DeBERTa, (b) ELECTRA, (c) DistilBERT, (d) RoBERTa

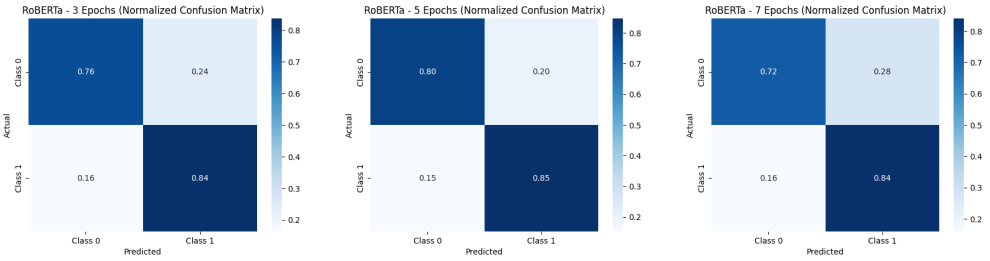

Fig. 5 Normalized confusion matrices for RoBERTa at 3, 5, and 7 epochs

4.3 Comparison

So if we talk about SVM first it does not perform well if we compare it with LLM and transformers.It happened because transformer models and LLM can understand the meaning of the sentences.After seeing the results we can say that the tranformers models can work better in understanding the complex conversation in movie like "Titanic".So the transformers models are very important for getting good and high accurate results.

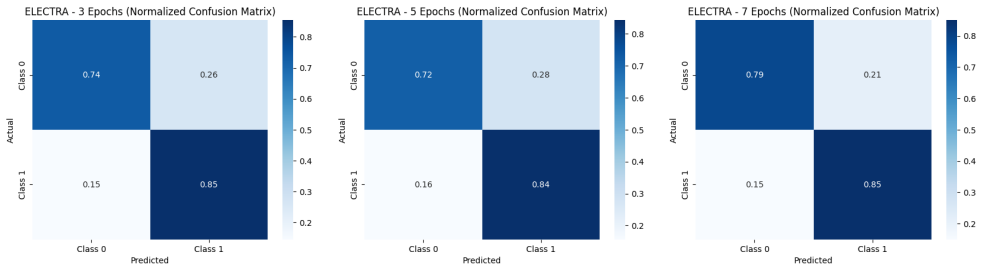

Fig. 6 Normalized confusion matrices for ELECTRA at 3, 5, and 7 epochs

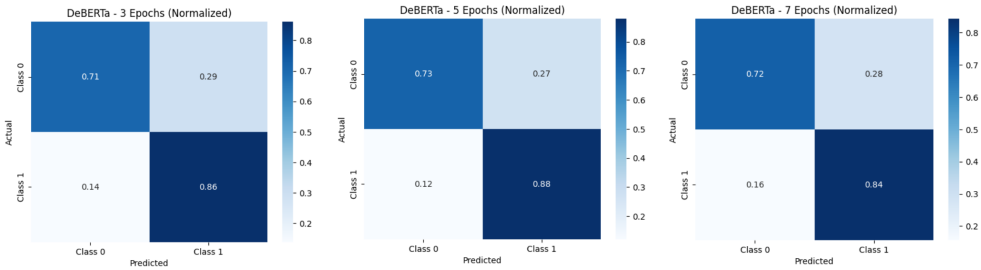

Fig. 7 Normalized confusion matrices for DEBERTA at 3, 5, and 7 epochs

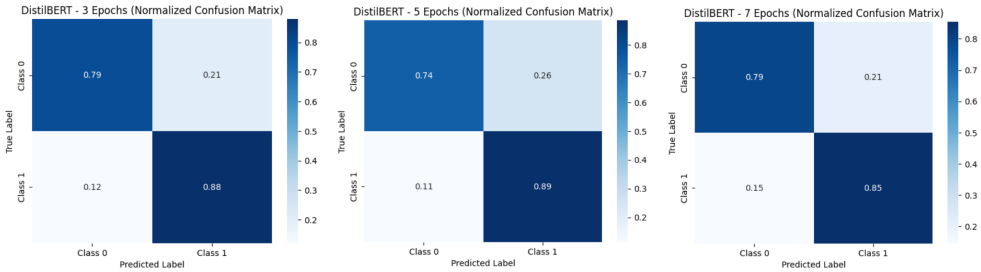

Fig. 8 Normalized confusion matrices for BART at 3, 5, and 7 epochs

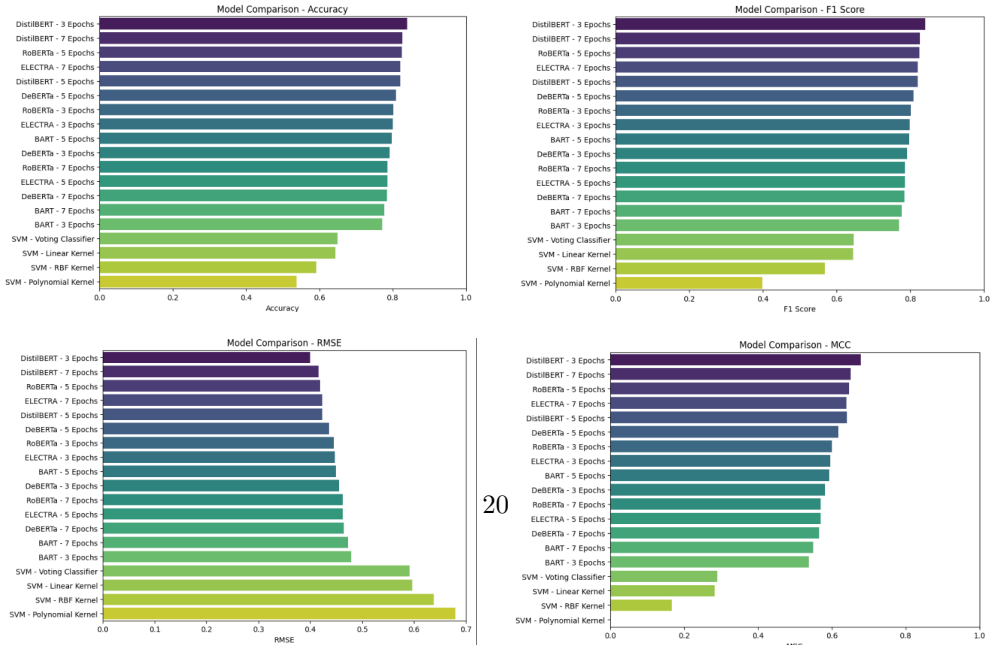

Fig. 9 Comparison of models: (a) Accuracy, (b) F1-score, (c) RMSE, (d) MCC

5 Conclusion

In this project we have first removed the noise, normalize texting from our dataset Titanic .We learnt some types of ML and Deep learning model and for the sentiment analysis of the dataset like Titanic movie which is very popular.So after evaluating and performing all models like Bart roberta electra distilbert n for trasformers we have trained SVm approaches also for better results.DistilBERT got the highest accuracy amond all of them.We got 84.0 accuracy.This study applied preprocessing techniques of noise removal and textual normalization to efficiently conduct sentiment analysis on a movie dataset. After this, transformer-based models including BART, RoBERTa, DeBERTa, DistilBERT, and ELECTRA were compared to traditional models, where SVMs with varying kernel functions were used. The analysis shows that especially for tasks in sentiment analysis, transformer models capture the semantics of context much better than classical models. DistilBERT was the most effective in reaching 84.0 for model accuracy, assuming the best trade-off of both speed and accuracy. These findings provide a good base for future models and demonstrate the advantage of extensive refinement as well as increased dataset size.

6 Future Work

Expansion of Dataset - We can train a larger dataset in the future so that the model can learn more patterns. This will reduce bias and improve overall performance.
Fine-grained Sentiment Analysis - Currently, we only classify sentiments as positive, negative, or neutral. In the future, we can extend this to emotions like sadness, fear, and frustration, which will help in deeper understanding.
Multimodal Sentiment Analysis - We can develop systems that analyze facial expressions, tone of speech, and background music or actions in real time to improve sentiment detection.
Explainable AI (XAI) - We can build models that explain which words or parts of a sentence influenced the prediction, making decisions more transparent.
Real-time Application - We can develop real-time sentiment analysis systems for live conversations or movie watching to provide instant emotional insights.

7 Social Impact

This research has some important effects on society.
For Social Media - Sentiment analysis can be used to detect if someone is posting something harmful on social media or engaging through likes and comments on posts.
For Better Customer Service - Businesses can use sentiment analysis to determine whether a customer is happy or not. If they are unhappy, companies can respond with better support or solutions, which helps in customer retention.
Mental Health Monitoring - For clinical purposes, doctors or nurses can use sentiment analysis to understand what a person is thinking based on messages, writings, or social media posts, and assess their mental condition.
Filmmaking - Filmmakers can use sentiment analysis for understanding emotional impact of scenes or dialogues to improve storytelling.

Educational Applications - Sentiment analysis can be used in education to understand students' emotions during online classes or discussions.

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